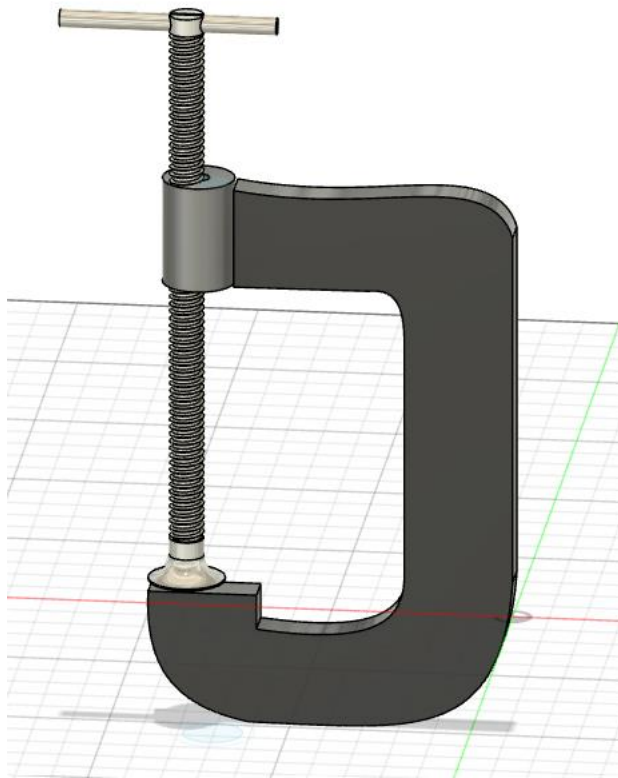
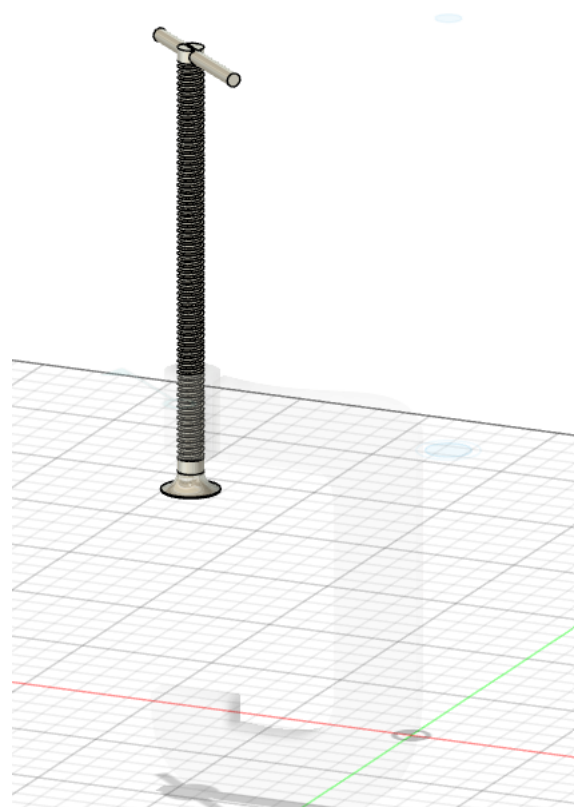
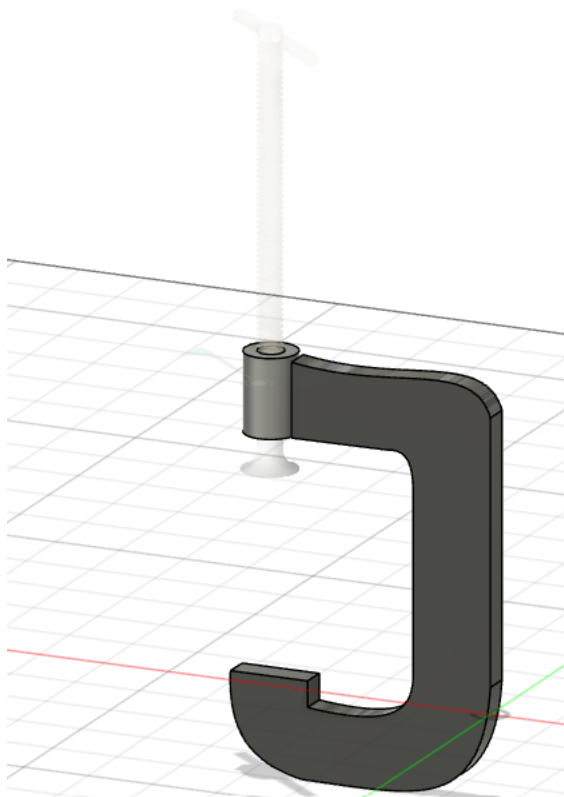


Engineering Design Workflow

Main Image

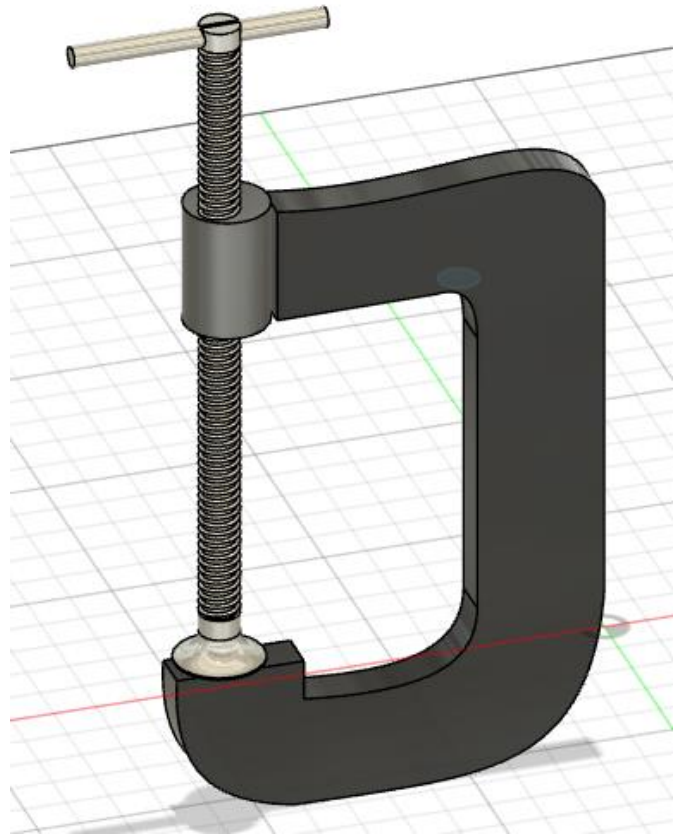
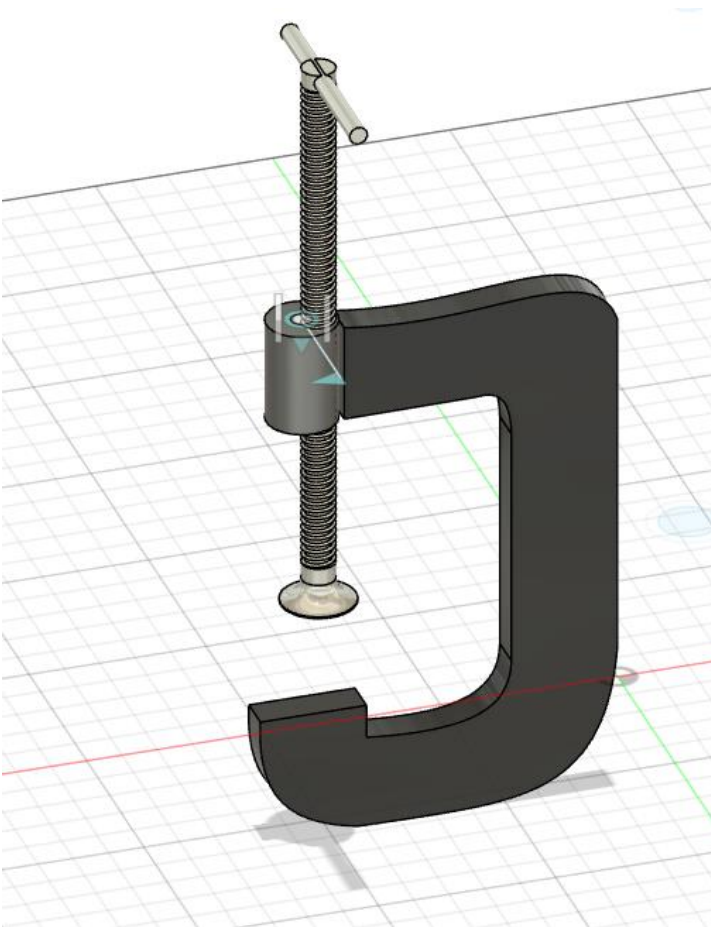


Modeled the main body and screw as independent components for assembly.



Joint Movement

Assembled the components and applied motion joints to simulate movement.



Engineering Challenges

Challenge	Resolution	Key Learning
Modeling the threaded screw mechanism	Simplified the screw geometry while preserving its functional appearance and assembly compatibility.	Effective engineering balances realism with design efficiency.
Structuring the assembly components	Created the frame, screw, handle, and swivel pad as independent components before assembly.	Proper component hierarchy simplifies assembly design and future modifications.
Aligning the screw with the frame	Corrected component positioning and joint alignment to ensure accurate rotational motion.	Precise alignment is essential for reliable mechanical assemblies.
Implementing rotational motion	Evaluated different joint configurations before applying a revolute joint for realistic screw rotation.	Selecting the correct joint type is critical for motion simulation.
Refining the overall design	Adjusted dimensions and component relationships to improve fit, appearance, and functionality.	Iterative refinement leads to more robust and manufacturable design